Journal of Food Composition and Analysis 123 (2023) 105501

Contents lists available at ScienceDirect

Journal of Food Composition and Analysis

journal homepage: www.elsevier.com/locate/jfca

Classification of fermented soybean pastes using inductively-coupled plasma optical emission spectroscopy and multivariate data modeling

Hyang Kim [a], Seongsoo Jeong [b], Hoeil Chung [b,*], Sang-Ho Nam [a,c,**], Yonghoon Lee [a,c,**]

[a] Department of Chemistry, Mokpo National University, Jeonnam 58554, Republic of Korea
[b] Department of Chemistry and Research Institute for Convergence of Basic Science, Hanyang University, Seoul 04763, Republic of Korea
[c] Plasma Spectroscopy Analysis Center, Mokpo National University, Jeonnam 58554, Republic of Korea

ARTICLE INFO

Keywords:
Fermented soybean paste
Inductively-coupled plasma optical emission spectroscopy
Classification modeling
Food analysis
Food composition

ABSTRACT

Fermented soybean paste is widely used in food sauces and stews in East Asia. The main ingredients are soybeans, salt, and water. Therefore, it provides both proteins and inorganic nutrients. Fermented soybean paste consumed in South Korea is produced in South Korea and China. In this study, for 166 fermented soybean pastes collected in Korean markets, nine inorganic elements were analyzed by inductively-coupled plasma optical emission spectroscopy (ICP-OES), and partial least squares-discriminant analysis (PLS-DA) models were developed to classify the samples with respect to their provenances. Among these elements, Mg and Sr were effective variables for discriminating between fermented soybean pastes produced in South Korea and those produced in China. The model trained using Mg and Sr exhibited a classification accuracy of 92.7%. Discrimination power of Mg was rationalized by the difference in Mg concentration between the salts used in South Korea (sea salts) and those used in China (refined salts). Our results indicated that the elemental content of fermented soybean paste was significantly influenced by the salts used as the main ingredients. The combination of ICP-OES and PLS-DA has been suggested as a reliable methodology for discriminating fermented soybean pastes produced in South Korea and those produced in China.

1. Introduction

Fermented soybean paste is one of the popular food sauces in East Asian countries such as China, Japan, and Korea (Shurtleff and Aoyagi, 1985; An et al., 2021). It is called *Doenjang* in Korea, *Miso* in Japan, and *Huangjiang* or *Dajiang* in China (Xie et al., 2019). The high content of both proteins and mineral elements in fermented soybean paste makes it a nutritionally excellent food (An et al., 2016). South Korea is an East Asian country in which various fermented soybean pastes are circulated in the market. Most of these are consumed in *Doenjang* stew and *Ssamjang*, a thick, spicy Korean paste made of *Doenjang*, red pepper paste, and other vegetables (Hwang and Kim, 2022; Kim et al., 2016). Fermented soybean pastes consumed in South Korean are currently produced in South Korea and China (Korea Agro-Fisheries and Food Trade Corporation, 2021). Fermented soybean paste is basically made of soybeans, salt, and water. However, the flavor and taste of fermented soybean paste vary depending on the production method and additional ingredients (Kim, et al., 2010). In addition to the main ingredients, some products contain additives such as rice, flour, sugar, and extracts from vegetables, seafood, and meat.

Although fermented soybean paste is a local food sauce popular in a few East Asian countries, its main ingredients come from various countries, such as South Korea, Australia, Vietnam, China, and United States. This causes the price of this popular food sauce to vary significantly, because companies located in South Korea and China have different supply chains for soybeans, salt, and other ingredients. Under these circumstances, a reliable methodology to distinguish between fermented soybean pastes produced in different countries would be helpful for regulating the fermented soybean paste market (Jeong et al., 2023). Based on the labels attached to the containers of fermented soybean paste products circulated in Korean markets, the soybeans and salt used for domestic fermented soybean pastes are mostly from South Korea. Particularly, the salts are solar sea salts produced on saltpans located along the west coast of the southern part of the Korean

Abbreviations: ICP-OES, Inductively-coupled plasma optical emission spectroscopy; PLS-DA, Partial least squares-discriminant analysis.

* Corresponding author.
** Corresponding authors at: Department of Chemistry, Mokpo National University, Jeonnam 58554, Republic of Korea.
E-mail addresses: hoeil@hanyang.ac.kr (H. Chung), shnam@mokpo.ac.kr (S.-H. Nam), yhlee@mokpo.ac.kr (Y. Lee).

https://doi.org/10.1016/j.jfca.2023.105501
Received 17 March 2023; Received in revised form 22 June 2023; Accepted 28 June 2023
Available online 29 June 2023
0889-1575/© 2023 Elsevier Inc. All rights reserved.

Peninsula. Those sea salts are famous for their high contents of mineral elements such as Mg, Ca, and K. (Kim, et al., 2019a; Kim, et al., 2019b; Park, et al., 2015). On the other hand, most fermented soybean pastes produced in China are made of soybeans from countries other than South Korea and refined salt or sea salts from Australia or Vietnam. The mineral contents of the salts are expected to be quite different from those of the Korean solar sea salts.

In this study, we analyzed Al, Ba, Ca, Mg, Sr, Cu, Fe, Mn, and Zn in 166 fermented soybean paste samples using inductively-coupled plasma optical emission spectroscopy (ICP-OES). Fermented soybean paste samples were collected from Korean markets. Of 166 samples, 101 were produced in South Korea. The others were produced in China and imported into South Korea. The capabilities of those elements to discriminate between the samples produced in South Korea and those imported from China were measured in terms of *interclass distance* defined as the difference between the two mean concentrations of Korean and Chinese samples scaled by a pooled standard deviation. Based on the interclass distances, Mg, Sr, and Zn were found to have effective discrimination capabilities. The discrimination capability of Mg was complementary to those of Sr and Zn, but the discrimination capabilities of Sr and Zn were not independent from each other. Therefore, the two sample classes, fermented soybean pastes produced in South Korea and those imported from China, were modeled by partial least squares-discriminant analysis (PLS-DA) based on the concentrations of Mg and Sr. For a separate test data set, the model employing Mg and Sr as latent variables provided a classification correctness of 92.7%. The concentration of Mg, the most effective latent variable in discriminating between the two classes, was found to be higher in the fermented soybean paste samples for which the Korean solar sea salts were used than those made of refined salts. Our results indicate that the concentrations of Mg and Sr can compose a chemical fingerprint that is useful for discriminating fermented soybean pastes produced in South Korea and those imported from China. Also, the combination of ICP-OES and PLS-DA is suggested as a dependable methodology for developing a classification model for screening particular fermented soybean paste products circulated in the markets.

2. Materials and Methods

2.1. Fermented soybean paste samples

The fermented soybean paste products were purchased from South Korean markets. All samples were produced by Korean companies. However, of the 166 collected samples, 101 were produced in South Korea and the other 65 were produced in China. Of the 101 samples produced in South Korea, 51 contained no additives other than the main ingredients, soybeans, salt, and water; the others contained additional ingredients such as herbs, red pepper, mushrooms, wheat, anchovy, shrimp, seaweeds, jujube, barley, potassium chloride, vinegar, citric acid, honey, ginseng, pollack, sweet pumpkin, and alcohol. Among the 65 samples produced in China, 53 included additional ingredients such as wheat, sugar, sodium glutamate, garlic, potassium sorbate, alcohol, mustard, sesame oil, rice, mushroom, anchovy, red pepper, mussel, clam, barley, yeast, katsuobushi, kelp, potassium iodide, onion, vinegar, shrimp, oyster, crab, beef, maltodextrin, chenpi, squid, stalked sea squirt, xanthan gum, vegetable extracts, and calcium lactate.

2.2. ICP-OES analysis

Each as-purchased sample was homogenized thoroughly, and a 10 g of it was taken for freeze drying. The homogenized samples were frozen at − 38 °C for 60 min and subsequently dried at 0 °C for 600 min and 35 °C for seven days. The freeze-dried samples were milled for 2 min using a food mixer (SGMF-660; Hanil Electric, Bucheon, South Korea). The sample for the ICP-OES analysis was prepared with the milled powder following Korean food standards codex (Korea Food & Drug Administration, 2023). A 1 g of each milled sample powder was placed in a high-pressure vessel, and then an 8 mL of nitric acid (70% aqueous solution; ODLAB, Kwangmyung, South Korea) and a 2 mL of hydrogen peroxide (30% aqueous solution; ODLAB, Kwangmyung, South Korea) were added. After 30 min for reaction, the mixture was further digested in a microwave oven (Microwave 5000; Anton Paar, Graz, Austria). By irradiating microwaves to the mixture, the temperature of the mixture was increased to 100 °C in 10 min, maintained at 100 °C for 2 min, further increased to 180 °C in 10 min, and maintained at 180 °C for 30 min. Then, the mixture was cooled at room temperature and stirred thoroughly using a vortex mixer (MaXshake TM VM30; DAIHAN Scientific, Wonju, South Korea). The resulting solution was filtered through a syringe filter (polyvinylidene fluoride, 0.45 μm; Thermo Scientific, Waltham, MA, USA). The filtrate was diluted to 250 mL. The dilution factor of this whole sample preparation process was 1/250. Standard solutions for calibration curves were prepared by diluting commercial stock solutions (Calib. Std #4 and #6; AccuStandard, New Haven, CT, USA) in the concentration ranges from 0.01 to 10 ppm for Al, Ba, Ca, Mg, Sr, Cu, Fe, Mn, and Zn. The sample and standard solutions were fed to the ICP-OES spectrometer (SPECTROGREEN; SPECTRO, Kleve, Germany) at the flow rate of 2 mL/min. The radio-frequency power was set to 1400 W. Agon was supplied for nebulizer, auxiliary, and coolant gases at the flow rates of 0.96, 1.0, and 13.0 L/min, respectively. To detect the analyte signals, the emission lines at 167.078 nm, 455.404 nm, 396.847 nm, 279.553 nm, 407.771 nm, 324.754 nm, 259.941 nm, 257.611 nm, and 213.856 nm were used for Al, Ba, Ca, Mg, Sr, Cu, Fe, Mn, and Zn, respectively.

Recovery efficiency of a few elements contained in the certified reference material (CRM) of soybean powder (KRISS CRM 108–01–005; Korea Research Institute of Standard Science, Daejeon, South Korea) was investigated for quality assurance of our ICP-OES analysis method. The soybean powder CRM contains Ca, Cu, Fe, and Zn, of which certified concentrations are 2725 ± 88, 11.71 ± 0.15, 69.1 ± 1.3, and 57.00 ± 0.71 mg/kg, respectively. A 1 g of the soybean powder CRM was digested and diluted by the same process as above, and the prepared solution was analyzed by the ICP-OES spectrometer. Calibration curves were also developed using the same standard solutions as those used for fermented soybean paste samples. As a result, the recovery efficiency values were 95.4%, 95.1%, 97.3%, and 94.0% for Ca, Cu, Fe, and Zn, respectively.

3. Results and discussion

3.1. Effective latent variables

The aim of this study was to develop a reliable statistical model to discriminate between fermented soybean pastes produced in South Korea and those produced in China based on the concentrations of the inorganic elements such as Al, Ba, Ca, Mg, Sr, Cu, Fe, Mn, and Zn. However, as mentioned in the Experimental section, many of the samples produced in South Korea and China contain various additional ingredients besides soybeans, salt, and water. Therefore, it should first be determined whether the additives affect the concentrations of the analyte elements. If the additives made significant contributions to the analyte concentrations and interfered with the correlation between the analyte concentrations and provenance (South Korea or China), the classification model based on those analyte concentrations would not be reliable.

Fig. 1 shows bar graphs comparing the concentrations of Al, Ba, Ca, Mg, Sr, Cu, Fe, Mn, and Zn in samples produced in South Korea with and without additional ingredients. As can be found in Fig. 1, the analyte concentrations in the samples with and without additives agreed well. It should be noted that the error bars in Fig. 1 represent not instrumental uncertainties but actual sample-to-sample concentration variations. This holds true for the results shown in Figs. 2 and 3. To measure the closeness between the two mean concentrations quantitatively, the difference between them was scaled using a pooled standard deviation

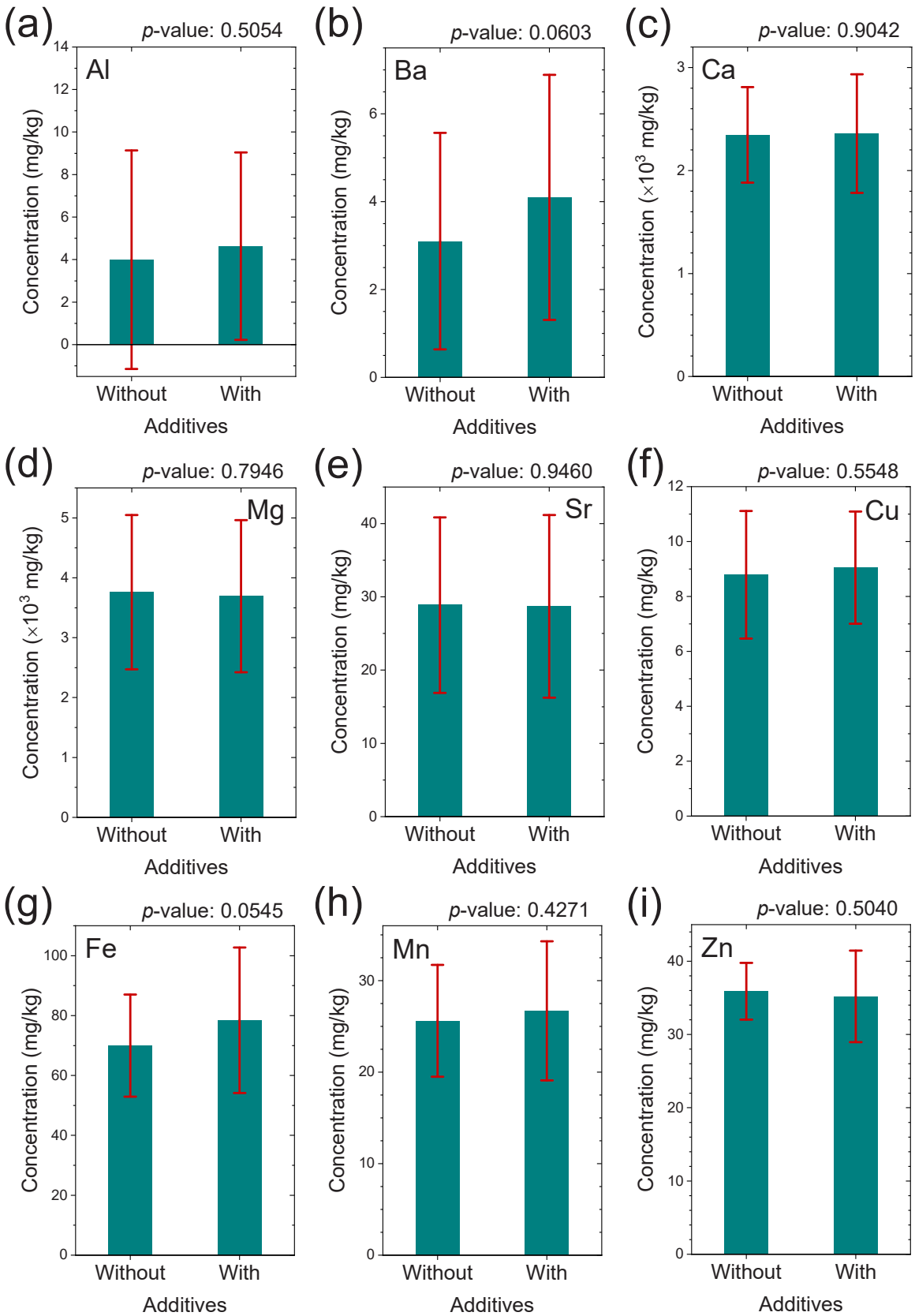

Fig. 1. Concentrations of (a) Al, (b) Ba, (c) Ca, (d) Mg, (e) Sr, (f) Cu, (g) Fe, (h) Mn, and (i) Zn obtained by ICP-OES for the soybean paste samples produced in South Korea with and without additives. p-values from ANOVA were noted along the corresponding panels.

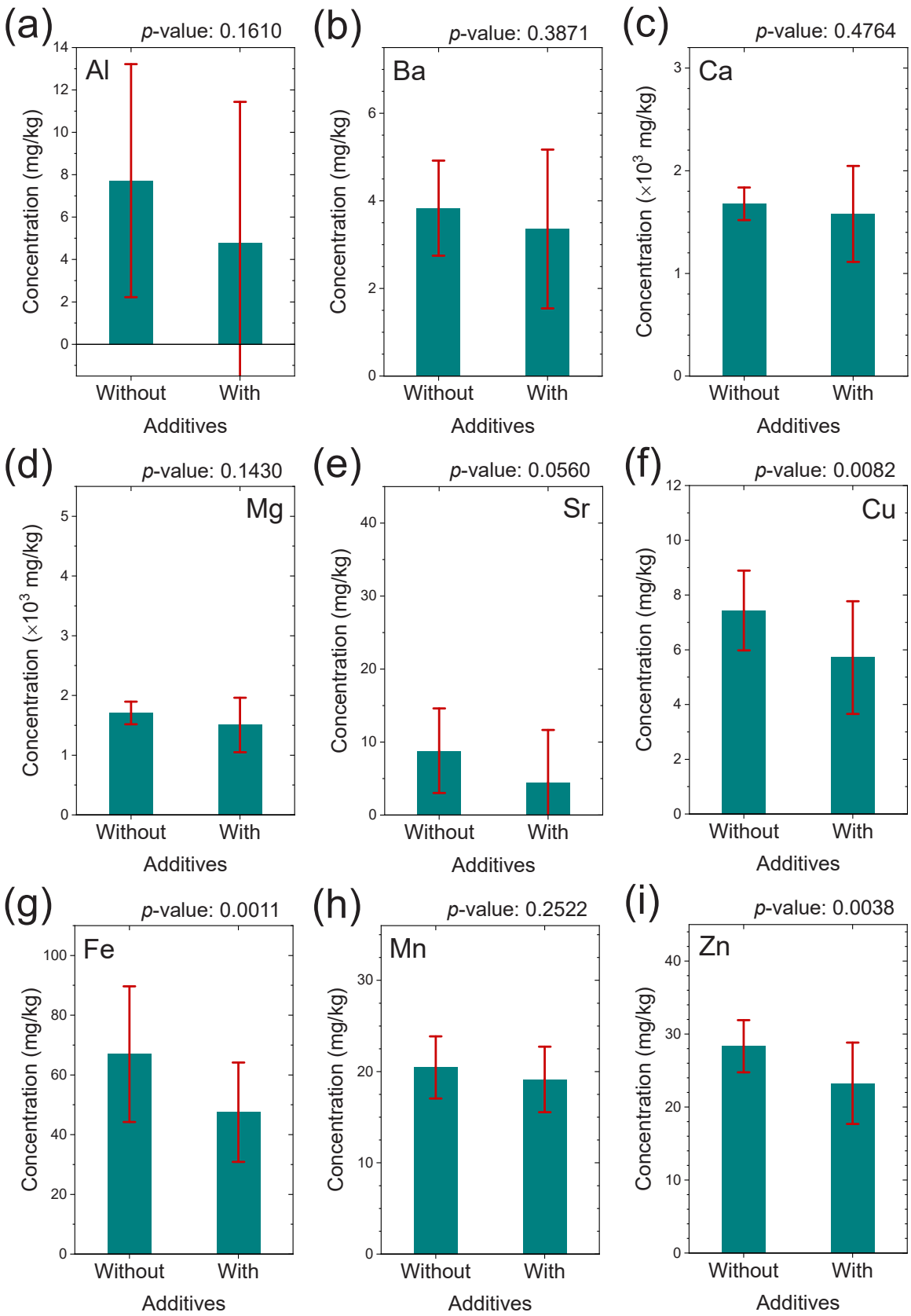

Fig. 2. Concentrations of (a) Al, (b) Ba, (c) Ca, (d) Mg, (e) Sr, (f) Cu, (g) Fe, (h) Mn, and (i) Zn obtained by ICP-OES for the soybean paste samples produced in China with and without additives. p-values from ANOVA were noted along the corresponding panels.

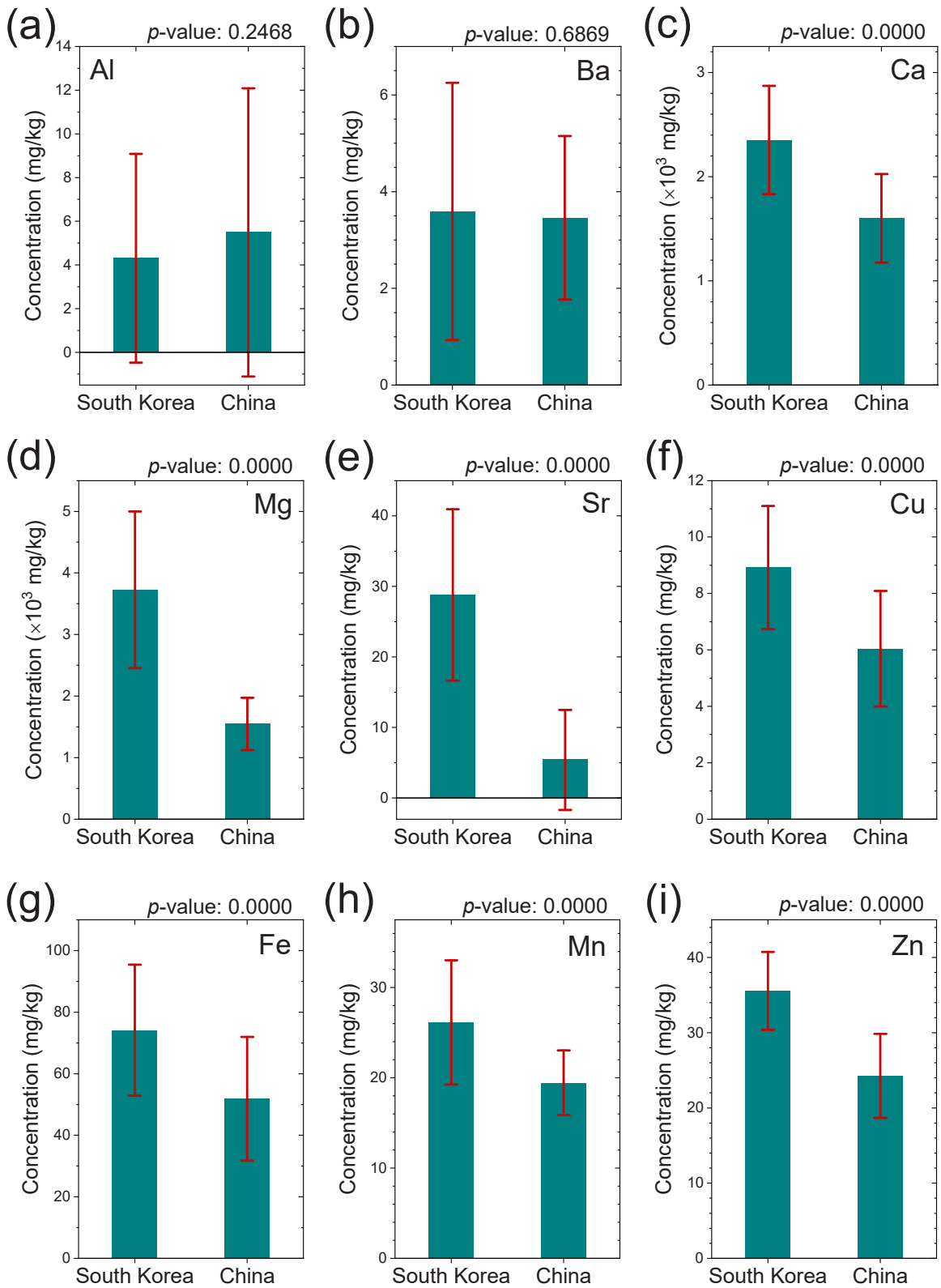

Fig. 3. Concentrations of (a) Al, (b) Ba, (c) Ca, (d) Mg, (e) Sr, (f) Cu, (g) Fe, (h) Mn, and (i) Zn obtained by ICP-OES for the soybean paste samples produced in South Korea and China. p-values from ANOVA were noted along the corresponding panels.

(International Union of Pure and Applied Chemistry, 2005; Park et al., 2022), which is the weighted average of the standard deviations of the two groups of concentrations obtained for the samples with and without additives. The pooled standard deviation, s_{pooled}, is expressed by Eq. (1).

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \tag{1}$$

In the above equation, n_1 and n_2 are the numbers of samples

belonging to group, 1 and 2, respectively. Herein, the two groups were the samples produced in South Korea *without* (group 1) and *with* (group 2) additives. So, n_1 is 51, and n_2 is 50. Also, s_1 and s_2 are the standard deviations of two groups 1 and 2, respectively. Thus, they correspond to the standard deviations of the concentrations measured for the samples produced in South Korea *without* and *with* additives, respectively. The s_{pooled} value was used to calculate the scaled difference, d_{scaled}, between the means of the group-1 and group-2 datasets, m_1 and m_2, respectively, using Eq. (2).

$$d_{scaled} = \frac{|m_1 - m_2|}{s_{pooled}} \tag{2}$$

The scaled differences were calculated to be 0.13, 0.38, 0.02, 0.05, 0.01, 0.12, 0.40, 0.16, and 0.13 for Al, Ba, Ca, Mg, Sr, Cu, Fe, Mn, and Zn, respectively, and all were much less than 1. This indicates that the concentrations analyzed for the samples with and without additives were very close to each other within $1 \times s_{pooled}$ The analysis of variance (ANOVA) was also performed with the intensity values from the two groups. As noted in panels of Fig. 1, all of the *p*-values were greater than 0.05. Thus, two mean concentrations of each element are in good agreement. This is consistent with the result based on the d_{scaled} values. Thus, for the samples produced in South Korea, the additives were found to make negligible contributions to the analyte concentrations. This also holds true for the samples produced in China.

Fig. 2 shows bar graphs comparing the concentrations of Al, Ba, Ca, Mg, Sr, Cu, Fe, Mn, and Zn in samples imported from China with and without additional ingredients. The scaled differences were obtained to be 0.45, 0.28, 0.23, 0.47, 0.62, 0.87, 1.09, 0.37, and 0.96 for Al, Ba, Ca, Mg, Sr, Cu, Fe, Mn, and Zn, respectively. Although these scaled differences were larger than those obtained for the samples produced in South Korea, they were still smaller than 1 except for that obtained for Fe. However, in the case of Fe, the scaled difference, 1.09, is very close to 1. Based on these results, it was found that the additives make no significant contribution to the analyte concentrations. The *p*-values from ANOVA are also noted along the corresponding panels in Fig. 2. For Al, Ba, Ca, Mg, Sr, Mn, the *p*-values are greater than 0.05. However, those of Cu, Fe, and Zn are a little less than 0.05. Thus, the two mean concentrations of each of these elements are statistically different. This implies that consideration of Cu, Fe, and Zn concentrations for the models discriminating fermented soybean paste samples would decrease the performance.

Fig. 3 shows bar graphs comparing the concentrations of Al, Ba, Ca, Mg, Sr, Cu, Fe, Mn, and Zn in samples produced in South Korea and China. For this comparison, the two kinds of samples with and without additives were not discriminated because the contributions of the additives were found to be negligible (see Figs. 1 and 2). The concentrations of Ca and Mg in the range of 1000 − 5000 ppm were particularly higher than those of the other elements. Al, Ba, and Cu were the elements contained in the fermented soybean pastes with the lowest concentrations ($\leq$ 10 ppm) among the nine elements. The concentrations of Fe, Mn, and Zn were at the level of several ten ppm. The concentration difference between the samples produced in South Korea and China was the largest for Sr. The concentration of Sr in the samples produced in South Korea was around 30 ppm and that in the samples imported from China was less than 10 ppm. Among the nine analyte elements, those showing the concentration difference between the two sample classes larger enough than the intraclass variation can be utilized for modeling the sample classes. To find out the effective latent variable, d_{scaled} values were calculated for the nine analyte elements. In this regard, n_1 and n_2 are the numbers of samples produced in South Korea and China, respectively, and s_1 and s_2 are the corresponding standard deviations. The s_{pooled} values calculated using Eq. (1) were employed to calculate the d_{scaled} values using Eq. (2) with m_1 and m_2, the mean concentrations in the samples produced in South Korea and China, respectively. The d_{scaled} values in this case can be regarded as the *interclass distance*.

Fig. 4 shows the d_{scaled} values of the nine analyte elements. Mg, Sr, and Zn exhibited d_{scaled} values greater than 2. This indicates that the three elements would have significant capabilities for discriminating between samples produced in South Korea and those from China. On the other hand, as shown in Figs. 3a and 3b, the concentrations of Al and Ba were almost the same in the two sample classes. This is reflected in their d_{scaled} values which are much smaller than 1 (0.21 and 0.06 for Al and Ba, respectively). This is in good agreement with the results from ANOVA. The p-values for the Al and Ba concentrations are remarkably greater than 0.05 (see the notes in Figs. 3a and 3b). This indicates that the two mean Al (or Ba) concentrations of the samples produced in South Korea and China are not different from each other. For the other elements, the p-values are much less than 0.00005. The d_{scaled} values of the other elements, Ca, Cu, Fe, and Mn, are 1.07 – 1.55. Thus, their discrimination capabilities would not be as effective as those of Mg, Sr, and Zn.

3.2. PLS-DA modeling

The two sample classes, fermented soybean pastes produced in South Korea and those imported from China, were modeled using PLS-DA (Barker & Rayens, 2003). The Unscrambler program (version 11.0, Camo Analytics) was used for the PLS-DA modeling. The concentrations of Mg, Sr, and Zn, that showed relatively large d_{scaled} values, were employed as latent variables, first.

Fig. 5 shows the loadings of the three major PLS factors. Factors 1, 2, and 3 explained 52.2%, 11.9%, and 1.8% of the data variation with sample classes, respectively. The loading of PLS factor 1 was dominated by the Mg concentration (Fig. 5a). This indicates that more than half of the discrimination power was provided by Mg. PLS factor 2, responsible for 11.9% of the data variation correlated with sample classes, was contributed by the concentrations of Sr and Zn (Fig. 5b). The contribution of Mg to PLS factor 2 is negligible. The loading values corresponding to Sr and Zn have the same signs. It means that the corresponding feature is the positive correlation between the concentrations of Zn and Sr. From the loadings of PLS factor 3, the corresponding feature is the negative correlation between the concentrations of Sr and Zn (Fig. 5c). The contribution of Mg to PLS factor 3 is also negligible. The signs of the loadings for Sr (+) and Zn (−) are different from each other. However, this feature explains a very small portion of the data variation (1.8%) correlated with the sample class.

Fig. 6 shows the PLS-factor score plots. As shown in Fig. 6a, the samples produced in South Korea show larger scores with respect to PLS factor 1 (black filled squares) than those produced in China (red open

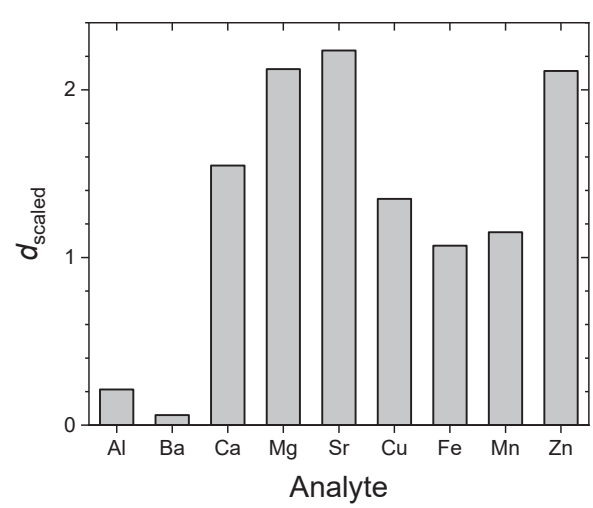

Fig. 4. Class distance along the concentrations of Al, Ba, Ca, Mg, Sr, Cu, Fe, Mn, and Zn.

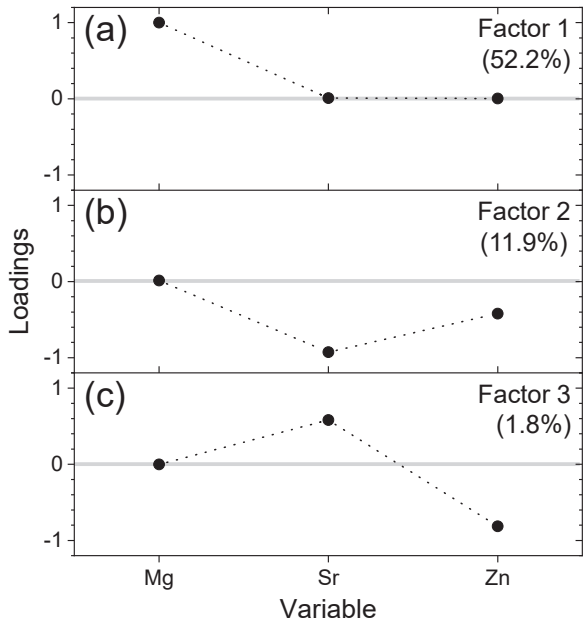

Fig. 5. PLS-factor loadings of the model based on the concentrations of Mg, Sr, and Zn.

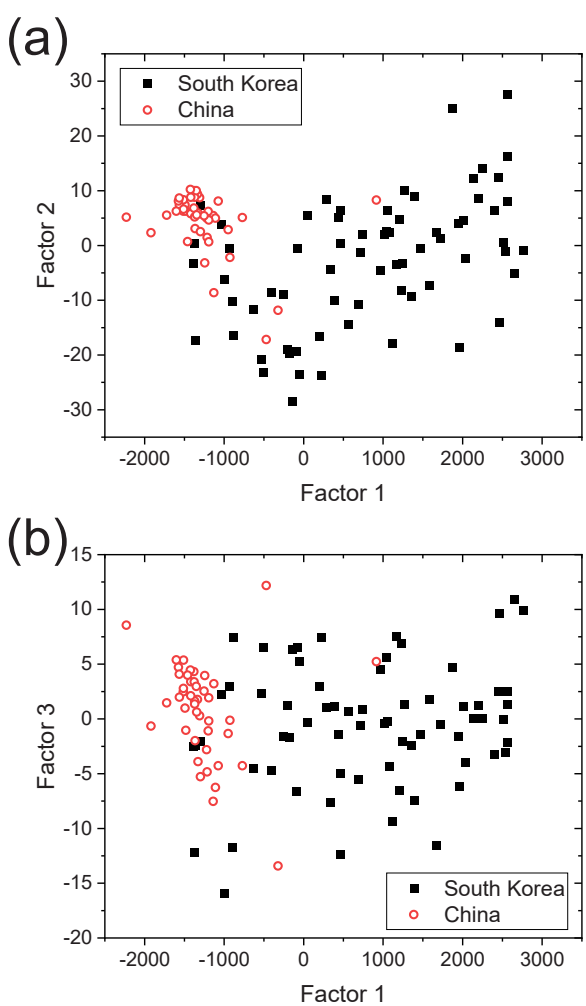

Fig. 6. PLS-factor score plots for the model based on the concentrations of Mg, Sr, and Zn.

circles). As mentioned above, the loadings of PLS factor 1 reflect the variation in the Mg concentration. Thus, the scores along the axis of PLS factor 1 indicates that the concentrations of Mg in the samples produced in South Korea is higher than those in the samples produced in China. This is well consistent with the results from the analysis of the Mg concentrations shown in Fig. 3d. The Mg concentrations in the samples produced in South Korea were 3700 ± 1300 ppm, and those in the samples produced in China were 1550 ± 430 ppm. The PLS factor 2 scores of the samples produced in China were slightly higher than those of the samples produced in South Korea. The loadings of PLS factor 2 explain the positively correlated concentrations of Sr and Zn, and the signs of both Sr and Zn concentrations are negative (Fig. 5b). Thus, the higher scores for the samples produced in China means the lower concentrations of Sr and Zn in them. This is also consistent with the results shown in Figs. 3e and 3i. Along the PLS factor 3 axis, the scores of the two sample classes were not separated from each other (Fig. 6b). This indicates that the discrimination capability of PLS factor 3 is negligible in comparison with those of PLS factors 1 and 2.

It should be noted again that (i) the concentrations of Sr and Zn are correlated with each other and (ii) Mg is decoupled from Sr and Zn. This indicates that the discrimination capabilities of Sr and Zn are not independent from each other and that one of the two correlated analytes, Sr and Zn, would be sufficient for PLS-DA modeling. The d_{scaled} value of Sr is slightly larger than that of Zn. Therefore, the simplest effective PLS-DA model can be developed using the two independent latent variables, Mg and Sr. To confirm this, four PLS-DA models were developed using different combinations of the analyte elements. Model 1 was the simplest model trained using the concentrations of Mg and Sr. Model 2 was trained using the concentrations of Mg, Sr, and Zn. Model 3 is based on the concentrations of Mg, Sr, Zn, and Ca. Ca is the element providing the fourth largest d_{scaled} as shown in Fig. 4. Model 4 was based on all the nine analyte elements. Among the data measured for 166 samples, those for 112 samples (68 from South Korea and 44 from China) were used to train the models, and the others for 55 samples (33 from South Korea and 22 from China) were used as a separate test data set that was used to validate the models. The classification correctness values of the four PLS-DA models are shown in Fig. 7a. The classification correctness decreased as the more analyte elements were considered. Model 1, trained using Mg and Sr, classified 92.7% ($= \frac{51}{55} \times 100\%$) of the test data correctly. The four misassigned test data correspond to the samples produced in South Korea. For all of the samples imported from China, Model 1 showed perfect classification accuracy. Model 2 was developed by adding the Zn concentration to the latent variables. However, the classification correctness (92.7%) is the same as that of Model 1. This is well consistent with the correlation between the Sr and Zn concentrations indicating that their discrimination capabilities are not independent. Models 3 and 4 showed classification correctness values of 90.9% ($= \frac{50}{55} \times 100\%$) and 85.5% ($= \frac{47}{55} \times 100\%$), respectively. The lower performances of Models 3 and 4, even with more information on elemental composition, can be attributed to the lower or no discrimination capabilities of Ca, Cu, Mn, Fe, Al, and Ba. These results from the PLS-DA modeling indicate that the two analyte elements, Mg and Sr, with independent discrimination capabilities, can train the classification model showing the best performance. The confusion matrices of Models 1 − 4 are also shown in Fig. 7b − 7e. From the confusion matrices, it was found that the classification correctness for the sample from China decreases as more variables are included in modeling. Based on the ANOVA results (see Fig. 2 f, 2 g, and 2i), this can be rationalized as following; when the concentrations of Cu, Fe, and Zn are included in the model, the fermented soybean paste samples without additional ingredients produced in China would be incorrectly classified as the samples produced in South Korea. The mean Cu, Fe, and Zn concentrations of the fermented soybean paste samples without additional ingredients produced in China are significantly higher than those with additional ingredients produced in China, and also closer to the

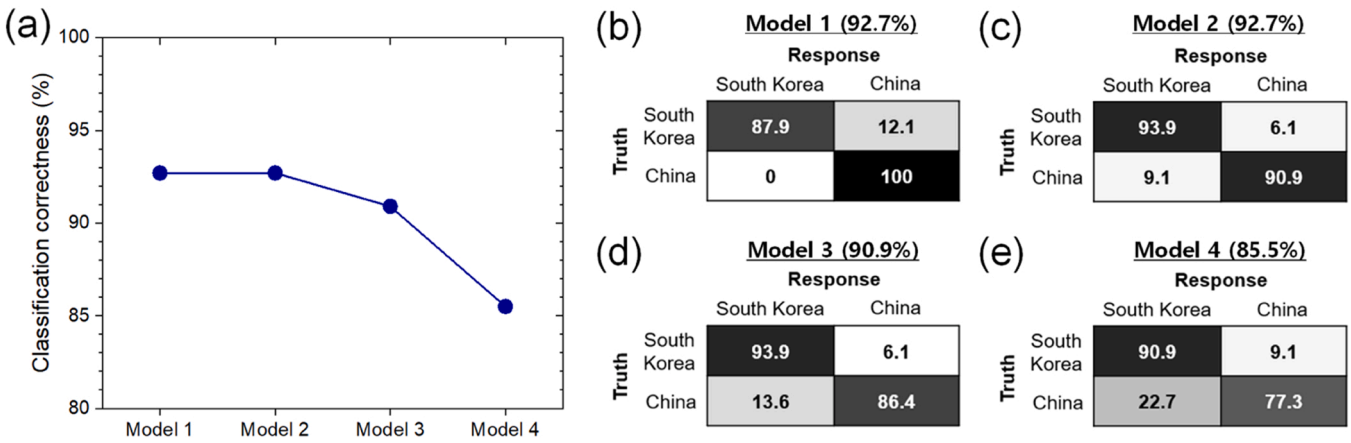

Fig. 7. (a) Classification correctness values of Models 1 – 4 and (b – e) confusion matrices of Models 1 – 4. In b – e, the correctness values are in percentage and the overall correctness value of each model are noted in the parentheses.

corresponding concentrations of the samples produced in South Korea.

The origin of the discrimination capability of Mg, responsible for the largest part of the data variation correlated with sample classes, was investigated. As shown in Figs. 1d and 2d, the additional ingredients made no significant contribution to the concentration of Mg. In addition, among the main ingredients (soybean, salt, and water), salt can introduce relatively large amounts of Mg into fermented soybean paste products. Thus, the difference in Mg concentration between the two sample classes (3750 mg Mg in 1 kg of the samples produced in South Korea – 1500 mg Mg in 1 kg of those produced in China = 2150 mg/kg) is mainly attributed to the elemental composition of the salts used for the production of the fermented soybean pastes. Fig. 8 shows pie charts presenting the sorts of salts used for the samples produced in South Korea and China. This information was collected from labels attached to the product containers. For the samples produced in South Korea, 55.4% ($=\frac{56}{101}$) of the samples used solar sea salts produced in South Korea, and 17.8% ($=\frac{18}{101}$) of the samples used solar sea salts without origin marks (Fig. 8a). Generally, solar sea salts contain larger amounts of Mg, Ca, and K than those in rock salts and refined salts, and particularly Korean solar sea salts are famous for the rich mineral content. At least 73.2% (= 55.4% + 17.8%) of the samples produced in South Korea were produced using solar sea salts. Thus, the high Mg concentrations in these samples can be attributed to the use of solar sea salts. On the other hand, it is found that refined salts were used in 73.8% ($=\frac{48}{65}$) of the samples produced in China (Fig. 8b). Refined salts are produced by removing other ions than Na^+ and Cl^- using chemical or physical processes (Zhang et al., 2021; Rathnayaka et al., 2014). Thus, the concentration of NaCl in refined salts is generally much higher than that in solar sea salts and consequently contains the smaller amounts of other mineral elements including Mg.

4. Conclusion

A combination of ICP-OES analysis and PLS-DA modeling is suggested as a dependable methodology for discriminating between fermented soybean pastes produced in South Korea and those produced in China. Among the analyzed elements, Mg, Sr, and Zn were found to have effective discrimination capabilities. However, the discrimination capabilities of Sr and Zn were not independent from each other. Thus, the simple but effective PLS-DA model could be developed based on the concentrations of Mg and Sr. Model performance was validated using a separate test data set and showed the classification correctness of 92.7%. The major source of Mg in fermented soybean pastes was identified as the salt, one of the main ingredients. The salts used to produce fermented soybean pastes were found to mainly influence the concentration of Mg. Based on the information obtained from the labels attached to the product containers, more than 73.2% of the fermented soybean paste samples produced in South Korea were made of solar sea salts which are famous for their high Mg concentration, whereas most of the salts used for producing the samples in China were identified as refined salts known to have low Mg concentration. This methodology, depending on the concentrations of Mg and Sr at the level of parts per million, is more practical than the state-of-the-art isotope analysis techniques in consideration of the required experimental skills, equipment complexity, and costs. Also, our results indicate that the identities of processed food products such as fermented soybean pastes, Kimchi,

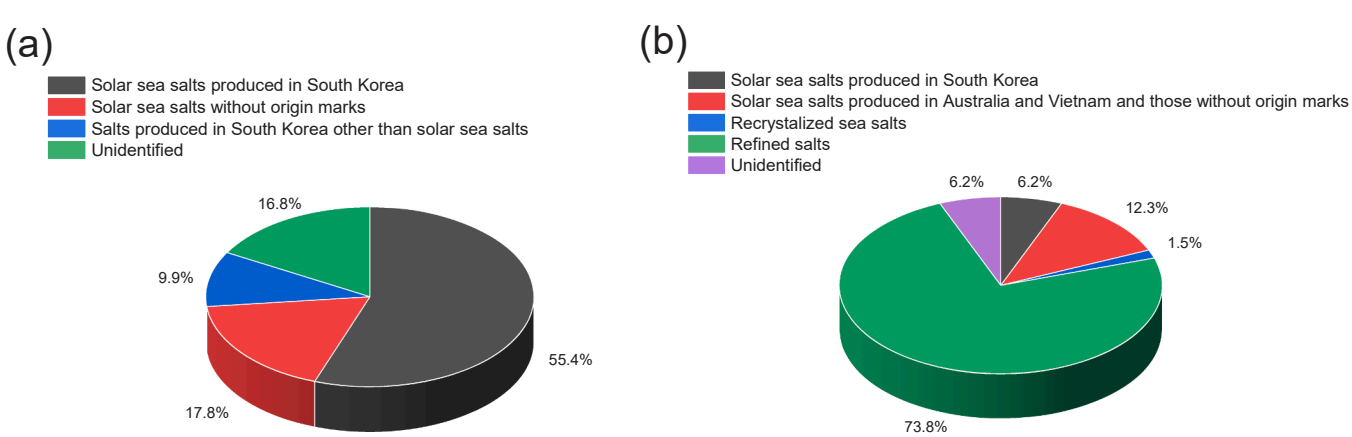

Fig. 8. Salts used to produce the fermented soybean pastes in (a) South Korea and (b) China.

etc. can be based on the provenances of their major ingredients. This emphasizes the importance of providing detailed information of ingredients used in processed food products.

Funding

This research was supported by Korea Basic Science Institute (KBSI) National Research Facilities & Equipment Center (NFEC) grant funded by the Korean Government (Ministry of Education) (No. 2019R1A6C1010005) and Basic Science Research Program through the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. 2022R1F1A1063812, No. 2020R1A6A1A06046728, and No. 2019R1A2C1003069).

CRediT authorship contribution statement

Hyang Kim: Formal analysis, Methodology, Validation. **Seongsoo Jeong:** Formal analysis, Methodology, Validation. **Hoeil Chung:** Conceptualization, Resources, Writing – review & editing. **Sang-Ho Nam:** Conceptualization, Resources, Validation, Writing – review & editing. **Yonghoon Lee:** Conceptualization, Resources, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgment

The authors are grateful to Dr. Jae-Min An (National Agricultural Products Quality Management Service, Republic of Korea) for providing fermented soybean paste products used in this work and Ms. Eunbi Kim for her technical assistant.

References

An, E.-J., Park, W.-J., Lee, J.-H., Kim, M., 2016. The nutritional analysis in fish/meat/doenjang and wild herbs/fish/meat Doenjang. Korean J. Food Nutr. 29 (*2*), 171–177. https://doi.org/10.9799/KSFAN.2016.29.2.171.

An, F., Li, M., Zhao, Y., Mu, D., Hu, X., You, S., Wu, J., Wu, R., 2021. Metatranscriptome-based investigation of flavor-producing core microbiota in different fermentation stages of dajiang, a traditional fermented soybean paste of Northeast China. Article 128509 Food Chem. 343. https://doi.org/10.1016/j.foodchem.2020.128509.

Barker, M., Rayens, W., 2003. Partial least squares for discrimination. J. Chemom. 17 (3), 166–173 https://doi:10.1002/cem.785.

Hwang, J., Kim, M.K., 2022. Effect of serving containers on the consumer acceptance of Doenjang stew. Article e12767 J. Sens. Stud. 37 (*4*). https://doi.org/10.1111/joss.12767.

International Union of Pure and Applied Chemistry. (2005). IUPAC Gold Book pooled standard deviation. Retrieved from https://goldbook.iupac.org/terms/view/P04758. Accessed February 26, 2023.

Jeong, S., Seol, D., Kim, H., Lee, Y., Nam, S.-H., An, J.-M., Chung, H., 2023. Cooperative combination of LIBS-based elemental analysis and near-infrared molecular fingerprinting for enhanced discrimination of geographical origin of soybean paste. Article 133956 Food Chem. 399. https://doi.org/10.1016/j.foodchem.2022.133956.

Kim, H., Jeon, Y., Lee, W.B., Nam, S.-H., Han, S.-H., Ham, K.-S., Singh, V.K., Lee, Y., 2019b. Feasibility of quantitative analysis of Mg and Ca in edible salts using a simple laser-induced breakdown spectroscopy device (https://doi:). Appl. Spectrosc. 73 (10), 1172–1182. https://doi.org/10.1177/0003702819861552.

Kim, H., Ngo, V.T., Kumar, S., Lee, W.B., Park, J., Han, S.-H., Nam, S.-H., Ham, K.-S., Lee, Y., 2019a. Laser-induced breakdown spectroscopy for rapid accurate analysis of Mg, Ca, and K in edible sea salts. Appl. Opt. 58 (*36*), 9940–9948. https://doi.org/10.1364/AO.58.009940.

Kim, H.G., Hong, J.H., Song, C.K., Shin, H.W., Kim, K.O., 2010. Sensory characteristics and consumer acceptability of fermented soybean paste (*Doenjang*). S375−S383 Food Sci. 75 (7). https://doi.org/10.1111/j.1750-3841.2010.01771.x.

Kim, S.Y., Park, B.R., Yoo, S.M., 2016. Quality characteristics of factory-style and handmade-style ssamjang. J. Korean Soc. Food Sci. Nutr. 45 (1), 100–108. https://doi.org/10.3746/jkfn.2016.45.1.100.

Korea Agro-Fisheries & Food Trade Corporation. (2021). Processed food market status report-soybean paste edition. Retrieved from https://www.atfis.or.kr/home/pdf/view.do?path=/board/202202/20220205113737447.pdf. Accessed February 26, 2023.

Korea Food & Drug Administration. (2023). Korean food standards codex. Retrieved from https://various.foodsafetykorea.go.kr/fsd/#/ext/Document/FC. Accessed June 20, 2023.

Park, G., Yoo, H., Gong, Y., Cui, S., Nam, S.-H., Ham, K.-S., Yoo, J., Han, S.-H., Lee, Y., 2015. Feasibility of rapid classification of edible salts by a compact low-cost laser-induced breakdown spectroscopy device. Bull. Korean Chem. Soc. 36, 189–197 https://doi.10.1002/bkcs.10043.

Park, J., Kumar, S., Han, S.-H., Singh, V.K., Nam, S.-H., Lee, Y., 2022. Two-step partial least squares-discriminant analysis modeling for accurate classification of edible sea salt products using laser-induced breakdown spectroscopy. Appl. Spectrosc. 76 (9), 1042–1050. https://doi.org/10.1177/00037028221091581.

Rathnayaka, D.D.T., Vidanage, P.W., Wasalathilake, K.C., Wickramasingha, H.W., Wijayarathne, U.P.L., Perera, S.A.S., 2014. Development of a process to manufacture high quality refined salt from crude solar salt. Int. J. Adv. Chem. Eng. Biol. Sci. 1, 137–142. https://doi.org/10.15242/IJACEBS.C0114189.

Shurtleff, W., & Aoyagi, A. (1985). The book of miso (2nd ed.). Ten Speed Press.

Xie, M., An, F., Yue, X., Liu, Y., Shi, H., Yang, M., Cao, X., Wu, J., Wu, R., 2019. Characterization and comparison of metaproteomes in traditional and commercial dajiang, a fermented soybean paste in northeast China. Food Chem.

Zhang, L., Zhang, Y., Hao, X., Ma, L., Gao, C., 2021. Preparation of deep-sea refined salt by solution crystallization technology. Article 042003. https://doi: IOP Conf. Ser.: Earth Environ. Sci. 804 https://doi.org/10.1088/1755-1315/804/4/042003.